GTC4Lusso USA - Tyres and rims - Latest Update: 2021-01-26

D2.02 Tyres and rims

Checking the state of the tyres

Tyre replacement is recommend after 4 years of normal use. Frequent use in maximum load conditions and at high temperatures may lead to premature deterioration.

Check the state of the tyres during the scheduled maintenance.

CAUTION

Pressure increases with increasing tyre temperature. Never reduce the tyre pressure if measured when the tyres are hot.

Tyre overheating due to underinflation may seriously compromise handling and tyre integrity.

Do not inflate the tyres to a pressure other than the specified value, as this would render the tyre pressure monitoring system useless.

After replacing one or more tyres, always calibrate the tyre pressure monitoring system by pressing the button on the passenger compartment dome light panel.

- Check the inflation pressure: pressure must be measured with cold tyres and must match the values specified.

CAUTION

Always inspect any tyre damage very carefully to assess severity: damage may greatly reduce tyre life and cause accidents.

- Check the tyres to ensure that there are no signs of damage such as abrasion, cuts, cracks, bubbles etc.
- Violent knocks against kerbs, holes or other obstacles and driving on poor road surfaces for extended periods may cause tyre damage. This damage may not always be visible. Any foreign objects penetrating the tyre in these conditions may cause structural damage that is only identifiable after removing the tyre from the wheel.

CAUTION

If the same tyres have been fitted on the vehicle for over 3 years, check frequently and carefully for signs of deterioration.

Do not monitor the pressure of a damaged tyre.

- Tyres will deteriorate even if the vehicle is used rarely or kept stationary. Cracks in the rubber on the tread and sidewalls, at times accompanied by blisters, are an indication of tyre deterioration.

CAUTION

Never fit a tyre with tread depth under the specified limit, even if only in a single localised area.

- Check the tread depth: the minimum permissible value is **1.7 mm**. This depth is identified by the tyre wear indicators.
- Check the wear over the entire tread; in the event of uneven tread wear, perform static and dynamic wheel balancing.

Note that once Run Flat tyres have been fitted, the instrument panel must be reprogrammed when refitting conventional tyres to prevent a warning message from being shown on the TFT display.

- Replace any tyres that have been punctured and/or been inflated with the tyre inflation bottle included with vehicle toolkit. For safety reasons, we do not recommend attempting to repair punctured tyres.

Preparation for tyre removal

The tyre pressure monitoring sensor is installed on the wheel rims, near the inflation valve.

Removing the tyre without using the correct tools and taking the necessary precautions may damage the sensor.

To avoid this problem, we are showing the entire procedure for removal and refitting of the tyres from the wheel rim, using a **CORGHI** tool, model **Artiglio A2019 RC** (shown in the photo) or **Artiglio A2020 TI**, recommended by Ferrari.

To prevent damage to the wheel rim, all parts of the tool that may come into contact with the rim are rubber coated.

Only use rubberised tools.
Wear gloves and protective eyewear.

Removing the tyre

Ensure that tyres are not kept in stock for more than 4 years. Tyres may be kept in stock for a maximum of 4 years, provided that they are kept out of direct sunlight, protected from the weather and humidity and stored in low oxygen conditions. If necessary, the Ferrari Service Network will verify whether old tyres are suitable for use. In all cases, tyres mounted on a vehicle for over 3 years must be inspected by an authorised Ferrari Service Centre.

Tyre replacement is recommend after 4 years of normal use. Frequent use in maximum load conditions and at high temperatures may lead to premature deterioration.

When removing and refitting tyres, DO NOT use tyre levers which could damage the rim and compromise its protective layer. Use levers coated in protective material.

The photos shown here were taken using a rim from the 575M Maranello.

To more clearly identify the correct mounting positions for the tools relative to the wheel, the vertical column of the tyre demounting machine is used as the starting point (12 o'clock).

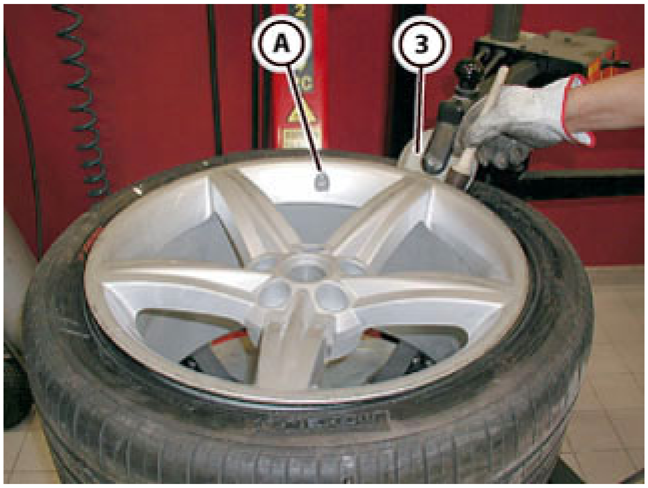
- Undo the core of the valve **(A)** to completely deflate the tyre.

- Place the wheel on the machine table and break the bead on both the inner and outer sides.
- Lubricate the bead **(1)** with the lubricant spray provided.

- Place the wheel on the self-centring clamp and use the bead breaker to ensure that the wheel is gripped correctly by the clamps **(2)** on the table.

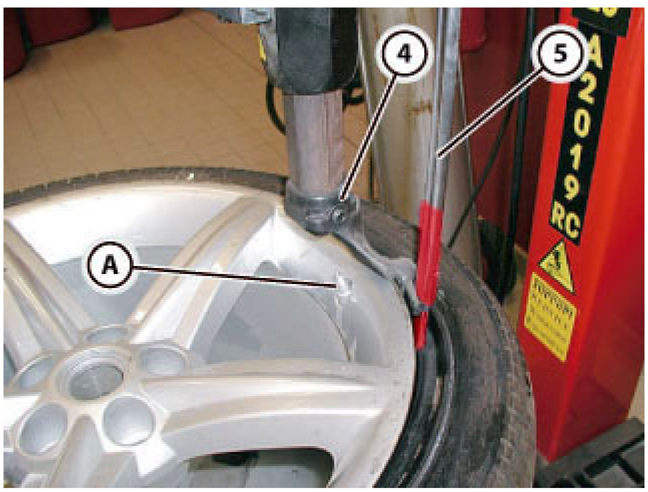
If the machine is not equipped with a bead breaker, clamp the wheel from the inside.

- Clamp the wheel.

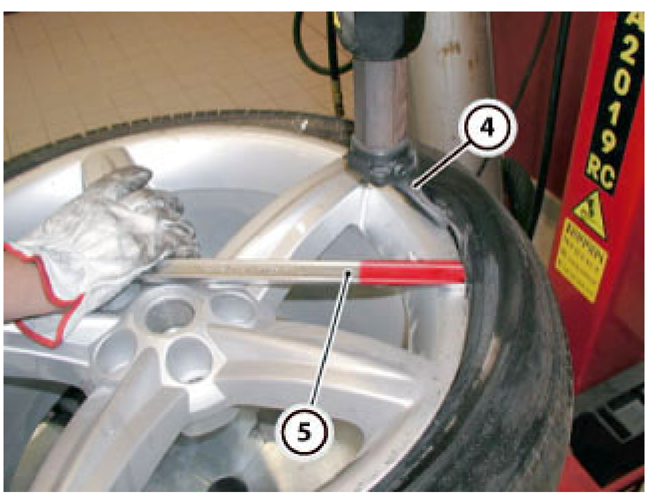


Observe the correct positions for bead breaking shown as follows.

- Position the inflation valve **(A)** at **12 o'clock**.
- Position the bead breaker disc **(3)**.
- Push the bead into the rim channel using the bead breaker disc while simultaneously turning the tyre counterclockwise, until the inflation valve **(A)** is at **3 o'clock**.
- Lubricate the bead while turning.
- To facilitate tyre removal, if necessary, insert a bead breaker lever at **180°** relative to the sensor.
- If the bead is not completely broken, repeat this stage in a clockwise direction, moving the inflation valve from **3 o'clock** to **12 o'clock**. Perform this stage repeatedly, clockwise and anticlockwise, until the bead is completely broken.



- Fit tool **(4)**: after clamping, the tool automatically moves upwards and outwards.
- Position the inflation valve **(A)** at **11 o'clock**.
- Mount the lever **(5)** on tool **(4)**, using the bead breaker disc to facilitate the operation if necessary.
- Remove the bead breaker disc.

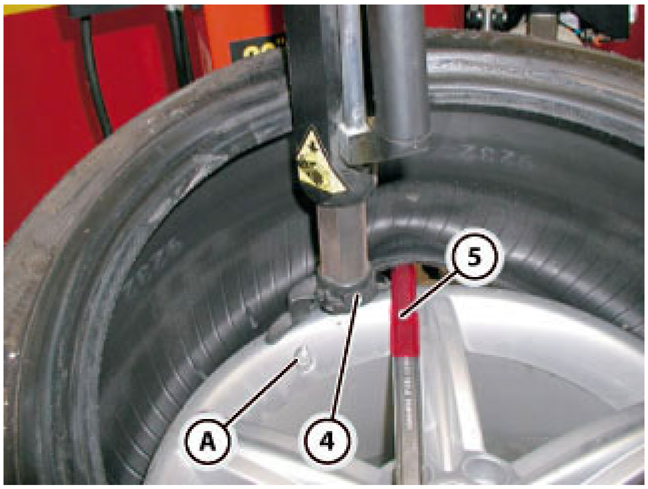


- Position the tyre bead on the tool **(4)** using the lever **(5)**.
- Remove the lever.

- Ensure that the entire edge of the bead is outside the rim and completely visible. If this is not the case, the bead may be damaged.

  If the bead does not slide easily into the groove, lubricate the contact area.

- Turn the wheel clockwise.



- Position the inflation valve **(A)** at **12 o'clock**.
- Fit the lever **(5)** on the tool **(4)**.
- Lift the tyre.

  Take care not to damage the sensor when inserting the tyre lever (5).

- Position the lower bead of the tyre on the tool **(4)**, using the lever **(5)**.
- Turn the wheel clockwise.

- Remove the rim from the machine and put carefully in a safe place.
- If the tyre is not to be refitted immediately, refit the valve core and cap.

➢ Remove the tyre pressure sensors (D2.03).

Proceed as follows if removal is necessary.

Checking wheel rims

- When carrying out regular maintenance or replacing tyres, check the state of the wheel rims.
- Visually inspect the external and internal wheel rim surfaces; if cracks or deep gouges are noted, replace the wheel.
- In the case of scratching, minor dents or superficial scuffs, repaint the affected area or, if necessary, the entire rim. Use the standard procedure for repainting an alloy wheel, applying a coat of silver based paint then, once the coat had dried to a matte finish, applying a layer of clear coat. Dry the wheel in the oven to polymerise the layers of paint applied.

Cover the valve mounting and the hub badge when repainting, as the additional thickness of the paint may impede correct assembly and compromise the effectiveness of the corrosion treatment applied.

- Take particular care around the mating surface with the brake disc; this area is only coated with a single layer of cataphoresis film.
- This treatment has a limited lifespan as the film is gradually eroded by fretting between the surfaces. This area cannot be repainted and a rim with a deteriorated film must be replaced.
- Before refitting, always check that the mating surfaces between the wheel and the brake disc are perfectly clean.

Using inappropriate tools for removing and refitting the tyre may damage the wheel rim and compromise the protective layer.

- Balance the wheel without the tyre but with the pressure sensor and valve installed: if, after removing the balancer weights, the values for static balance exceed **36** (including vertical bouncing and wobble), the rim must be replaced.

Refitting the tyre

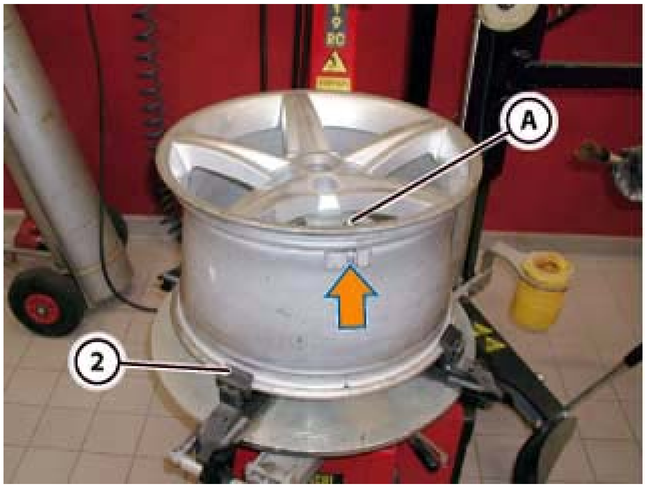
Ensure that tyres are not kept in stock for more than 4 years. Tyres may be kept in stock for a maximum of 4 years, provided that they are kept out of direct sunlight, protected from the weather and humidity and stored in low oxygen conditions. If necessary, the Ferrari Service Network will verify whether old tyres are suitable for use. In all cases, tyres mounted on a vehicle for over 3 years must be inspected by an authorised Ferrari Service Centre.

Tyre replacement is recommend after 4 years of normal use. Frequent use in maximum load conditions and at high temperatures may lead to premature deterioration.

When removing and refitting tyres, DO NOT use tyre levers which could damage the rim and compromise its protective layer. Use levers coated in protective material.

➤ Refit the tyre pressure sensors (D2.03).

Proceed as follows if the component has already been removed.



- After installing the pressure sensor **(A)**, mount the clean rim on the self-centring clamp, ensuring that the wheel is gripped correctly by the clamps **(2)** on the table.
- Lubricate the wheel.

- Pick up the tyre and ensure that the direction of rotation is correct.
- Lubricate both beads.

- Fit the inflation valve **(A)** at **6 o'clock**.
- Place the tyre on the rim and tip the tool arm **(4)** forwards.
- Fit the lower bead under the right hand part of the tool **(4)**.
- Turn the wheel clockwise.
- Mount the bead on the rim, applying constant pressure.
- Turn the wheel, bringing the inflation valve **(A)** to **12 o'clock**.

The bead breaker disc must never reach the valve-sensor assembly position.

- Fit the inflation valve **(A)** at **4-5 o'clock**.
- Place the tyre on the rim, matching the lighter parts of the rim, marked with an embossed letter **"G"**, with the yellow dots on the tyre.
- Insert the lever **(5)** at **2-3 o'clock**, using the bead breaker disc **(3)** to push the bead in.

If available, use tyre pliers instead of the lever **(5)**, with the appropriate adapter fitted.

If necessary, insert a second set of tyre pliers, with the correct adapter fitted.

- Turn the wheel clockwise until the upper bead is completely mounted.
- If used, remove the tyre lever with the bead breaker disc.

- Release the wheel from the clamps **(2)** on the table.

- Refit the core of the valve **(A)**.

- To ensure that the bead is mounted correctly, inflate the tyre immediately, while the beads are still lubricated.

The tyre must be inflated to **activate** the tyre pressure monitoring system sensor. Inflate the tyre to a pressure greater than **0.6 bar and keep inflated at this pressure for a few minutes**.

- If the maximum permissible inflation pressure of **(3.2 bar / 44 psi)** is not sufficient to seat the beads, lubricate again.

➤ Balance the wheels (D2.02).

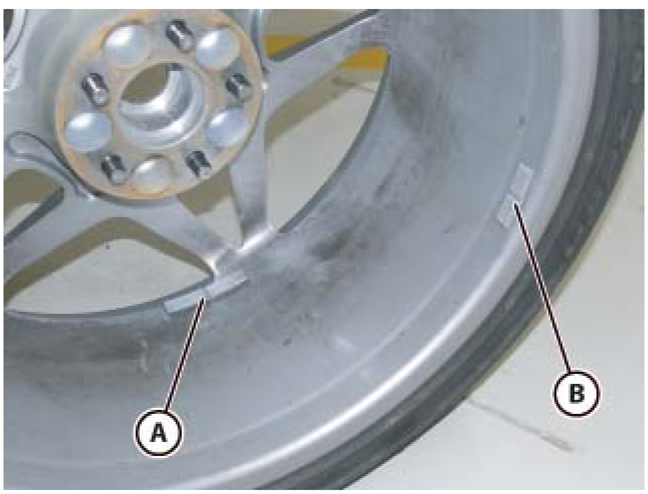
Wheel balancing

Only use self-adhesive counterweights. In addition to damaging the wheel rim, other types of counterweights can interfere with parts of the suspension or detach while driving.

Do not touch the adhesive surface of the counterweight before applying to the rim and never reuse counterweights or use weights with no protective paper, as these may detach while driving.

After replacing a tyre, wheel rim, or both, the wheel must be statically and dynamically balanced using a balancing machine and suitable counterweights.

Manufacturers identify the heaviest part of the wheel rim and the tyre with a reference mark. When mounting the tyre onto the rim, always set these reference marks opposite each other to minimise the amount of counterweights necessary for balancing.



- The counterweights must be applied in position **(A)** on the inner edge of the wheel rim, in position **(B)**, or in both positions if necessary.
- Thoroughly clean the wheel rim surface onto which the counterweight will be applied with heptane.
- Remove the protective backing and affix the counterweight onto the wheel rim, pressing evenly to ensure that it adheres perfectly.